Module 1 Homework 1 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:18

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

Which of these fractions is the biggest?

- $\frac{1}{2}$
- $\frac{2}{3}$
- $\frac{3}{4}$
- $\frac{4}{5}$
- $\frac{5}{6}$

2

Review Later

Which of these fractions is the biggest?

- $\frac{6}{5}$
- $\frac{5}{4}$
- $\frac{4}{3}$
- $\frac{3}{2}$
- $\frac{2}{1}$

3

Review Later

Which of the following decimals is equal to $\frac{3}{7}$?

- $0.\overline{428571\ldots}$
- $0.\overline{285714\ldots}$
- $0.\overline{857142\ldots}$
- $0.\overline{571428\ldots}$

- $0.\overline{714285\ldots}$

4

Review Later

Find the value of

$$\frac{14}{15} \times \frac{12}{13} \times \frac{16}{17} \times \frac{13}{14} \times \frac{15}{16}.$$

Your answer should be a fraction in simplest form written without spaces, e.g. 1/2.

5

Review Later

If your basketball shooting success rate is $\frac{1}{3}$, then which of the following is true?

- You made one miss shot and three successful shots
- You made one successful shot and three miss shots
- You made one successful shot, and then three more successful shots
- You made one miss shot, and three times as many successful shots
- You made one success shot, and three times as many total shots (misses plus successes)

6

Review Later

Find which of the following is equal to

$$0.1\overline{666} \times 12$$

- $\frac{1}{6}$
- $\frac{4}{3}$
- 2
- 4

0.22

7

Review Later

What is $1.125 \times 2.\overline{666}$?

8

Review Later

Find the value of

$$\left(\frac{1}{1+\frac{1}{3}}\right) \times \left(\frac{1}{1+\frac{1}{4}}\right) \times \left(\frac{1}{1+\frac{1}{5}}\right) \times \left(\frac{1}{1+\frac{1}{6}}\right)$$

Your answer should be a fraction written without spaces, like this: 1/2

9

Review Later

While using a computer program, you accidentally scaled a photograph down to 80% of its previous size, and saved, so you can't undo your mistake. By what percent should you increase the dimensions of the photograph in order to undo your mistake?

- 25%
- 20%
- 15%
- 10%
- 5%

10

Review Later

Over summer you visit your grandmother, and return home to find that you've gained weight; 11% of your weight, to be precise. Shortly after, you eat some mysterious cafeteria food and fall sick for several weeks, which makes you lose 11% of your current weight. How does your weight now compare to your weight before summer?

- Your weight now is the same as your weight before summer.
- Your weight now is up by 2.1%
- Your weight now is down by 1.21%
- Your weight now is up by 10%
- Your weight now is down by 1%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

Baby Alice spends $\frac{5}{8}$ of the day sleeping, $\frac{5}{24}$ of the day eating, and the rest of the day playing. Baby Bob spends $\frac{7}{12}$ of the day sleeping, $\frac{1}{6}$ of the day eating, and the rest of the day playing. How many more hours a day does Bob spend playing compared to Alice?

12

Review Later

Einstein was once quoted as saying, "Half of my students study math, one-third of my students study geometry, one-seventh study chemistry, and 20 of them don't study at all." (For the purposes of this question, assume that math does not include geometry.) How many students did Einstein have?

13

Review Later

Find the last digit of $(\frac{5}{8})^5$.

14

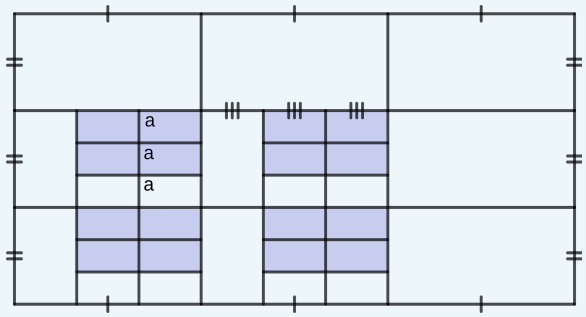
Review Later

Calculate

$$33\left(3+\frac{1}{3+\frac{1}{3+\frac{1}{3}}}\right)$$

15

Review Later



What fraction of the rectangle above is colored white? Your answer should be a fraction without spaces, e.g. 1/2.

16

Review Later

Which of the following fractions is the largest?

- $\frac{4}{9}$
- $\frac{50}{101}$
- $\frac{49}{99}$
- $\frac{9}{19}$
- $\frac{99}{200}$

17

Review Later

Your initial basketball shot success rate is $\frac{1}{12}$, but if you make four more consecutive shots (without any misses), then your success rate will become $\frac{3}{14}$. How many misses did you make so far?

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

Which of the following fractions is the largest?

- $\frac{1110}{1111}$
- $\frac{2223}{2225}$
- $\frac{3334}{3337}$

19

Review Later

Repeat the following steps, starting with an equilateral triangle: connect the midpoints of the three sides to form four smaller equilateral triangles. Color the innermost triangle black. Repeat this process with all remaining white equilateral triangles for a certain number of times, n, until the fraction of the large triangle that is shaded black is $\frac{37}{64}$. (The first step should be included in the value of n.)

What is the value of n?

20

Review Later

You've made 8 liters of lemonade that is 50% water and 50% lemonade mix. However, it is much too sweet for you, and you think the concentration of lemonade mix needs to be halved to 25% for optimal flavor. The only issue is, you have no pure water! The only other liquid you have on hand is another more diluted pitcher of lemonade that is 20% lemonade mix. How many liters of the diluted lemonade must be added to your original 8 in order to get the desired concentration?

View Marked Questions

Finished

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